Algorithm S1 Phase 1 — Static Surface Scan

Require: Robot positioned at start of stroke path ($y = 0$).
Ensure: Initial angle map $\boldsymbol{\theta}^{(0)}_{\text{fwd}}$, $\boldsymbol{\theta}^{(0)}_{\text{bwd}}$.

```
    — Forward scan (distal → proximal) —
for i = 0 to N−1 do
    Retract end-effector to non-contact position
    Move robot to measurement point y_i; stop
    if i = 0 then                                   ▷ First point: loading acquisition
        θ ← LoadingAcq(F*, ε_tol)
    else                                            ▷ Subsequent points: unloading acquisition
        θ ← UnloadingAcq(F*, ε_tol)
    end if
    θ^(0)_fwd(y_i) ← θ
end for

    — Backward scan (proximal → distal) —
for i = 0 to N−1 do                                 ▷ Points in reverse stroke order
    Retract end-effector to non-contact position
    Move robot to measurement point y_i; stop
    θ ← LoadingAcq(F*, ε_tol)
    θ^(0)_bwd(y_i) ← θ
end for

Linearly interpolate between measured points to obtain dense maps
return θ^(0)_fwd, θ^(0)_bwd
```

```
function UnloadingAcq(F*, ε_tol)
    Lower end-effector rapidly until F ≥ F_over                ▷ F_over = 0.7 N
    repeat
        Raise by Δθ_step; wait for settling                    ▷ Δθ_step = 1.5°
        Read force F
    until |F − F*| ≤ ε_tol
    return current motor angle θ
end function

function LoadingAcq(F*, ε_tol)
    Lower end-effector rapidly until contact (F ≥ F_contact)   ▷ F_contact = 0.1 N
    repeat
        Lower by Δθ_step; wait for settling
        Read force F
    until |F − F*| ≤ ε_tol or F > F* + ε_tol
    return current motor angle θ
end function
```

Algorithm S2 Phase 2a — Dynamic Learning via Iterative Refinement

Require: Static angle maps $\boldsymbol{\theta}_{\text{fwd}}^{(0)}$, $\boldsymbol{\theta}_{\text{bwd}}^{(0)}$ from Algorithm S1.
Ensure: Learned angle maps $\boldsymbol{\theta}_{\text{fwd}}^{(K)}$, $\boldsymbol{\theta}_{\text{bwd}}^{(K)}$.

for $k = 1$ **to** K **do** ▷ $K = 5$ trials
  Reset to start; lower to $\theta_{\text{fwd}}^{(k-1)}(y_0)$; warm up PID
  — **Forward stroke** —
  Execute stroke: $\theta_{\text{cmd}} = \theta_{\text{fwd}}^{(k-1)}(y) - u_{\text{PID}}$
  Record $u_{\text{PID}}(y_i)$, $e(y_i) = F^* - F(y_i)$ at each point y_i
  — **Backward stroke** —
  Execute stroke: $\theta_{\text{cmd}} = \theta_{\text{bwd}}^{(k-1)}(y) - u_{\text{PID}}$
  Record $u_{\text{PID}}(y_i)$, $e(y_i)$ at each point y_i
  — **Update rule (fwd / bwd independently)** —
  for $i = 0$ **to** $N-1$ **do**
    if $|e(y_i)| < \delta_{\text{dz}}$ **then** $\theta^{(k)}(y_i) \leftarrow \theta^{(k-1)}(y_i)$ ▷ Dead zone
    else $\theta^{(k)}(y_i) \leftarrow \theta^{(k-1)}(y_i) + \alpha \cdot u_{\text{PID}}(y_i)$
    end if
  end for
  MonotonicityFilter($\boldsymbol{\theta}^{(k)}$, $\mathbf{e}$) ▷ Alg. S3
end for
return $\boldsymbol{\theta}_{\text{fwd}}^{(K)}$, $\boldsymbol{\theta}_{\text{bwd}}^{(K)}$

Algorithm S3 Phase 2b — Monotonicity Filter

Require: Updated angle map $\boldsymbol{\theta}$, error vector $\mathbf{e}$.
Ensure: Monotonicity-corrected $\boldsymbol{\theta}$.

— **Ceiling check (distal → proximal)** —
for $i = N-2$ **downto** 0 **do**
  if $|e(y_i)| \geq \delta_{\text{dz}}$ **and** $\theta(y_i) > \theta(y_{i+1})$ **and** $|e(y_{i+1})| < \delta_{\text{trust}}$ **then**
    $\theta(y_i) \leftarrow \theta(y_{i+1})$ ▷ Clamp to neighbor
  end if
end for
— **Floor check (proximal → distal)** —
for $i = 1$ **to** $N-1$ **do**
  if $|e(y_i)| \geq \delta_{\text{dz}}$ **and** $\theta(y_i) < \theta(y_{i-1})$ **then**
    if $|e(y_i)| < \delta_{\text{trust}}$ **and** $|e(y_{i-1})| \geq \delta_{\text{trust}}$ **then**
      $\theta(y_{i-1}) \leftarrow \theta(y_i)$ ▷ Pull untrusted neighbor
    else
      $\theta(y_i) \leftarrow \theta(y_{i-1})$ ▷ Clamp current point
    end if
  end if
end for
return $\boldsymbol{\theta}$

Algorithm S4 Real-Time FF+PID Force Control During Experiment

Require: Learned angle maps $\boldsymbol{\theta}_{\text{fwd}}$, $\boldsymbol{\theta}_{\text{bwd}}$ from Algorithm S2; PID gains $K_p = 0.3$, $K_i = 0.03$, $K_d = 0.03$.
Ensure: Force-controlled stroking at target $F^* = 0.4\,\text{N}$.

Initialization:
Tare load cell; calibrate odometry sensor
Slowly lower end-effector to $\theta_{\text{fwd}}(y_0)$
Warm up PID for 5 s (stationary, with elevated K_i)
Reset integral term; switch to nominal K_i

for each stroke cycle $= 1$ **to** C **do** ▷ $C = 10$ round trips
  — **Forward stroke** —
  while $y < y_{\max}$ **do** ▷ $y_{\max} = 120\,\text{mm}$
    $F \leftarrow \textsc{ReadForce}(\ )$ ▷ 5-sample moving average, 20 ms interval
    $e \leftarrow F^* - F$
    $u_{\text{PID}} \leftarrow K_p\, e + K_i \int e\, dt + K_d\, \dot{e}$ ▷ Updated every 50 ms
    $\theta_{\text{FF}} \leftarrow \textsc{Interpolate}(\boldsymbol{\theta}_{\text{fwd}},\ y)$ ▷ Linear interpolation
    $\theta_{\text{cmd}} \leftarrow \theta_{\text{FF}} - u_{\text{PID}}$
    Apply θ_{cmd} to motor (clamped to safe range)
    Log $\{t,\ F,\ y,\ x,\ \psi,\ v\}$ ▷ Every 100 ms
  end while

  — **Direction reversal** —
  Stop robot; reset PID integral term

  — **Backward stroke** —
  while $y > y_{\min}$ **do** ▷ $y_{\min} = 0$
    $F \leftarrow \textsc{ReadForce}(\ )$
    $e \leftarrow F^* - F$
    $u_{\text{PID}} \leftarrow K_p\, e + K_i \int e\, dt + K_d\, \dot{e}$
    $\theta_{\text{FF}} \leftarrow \textsc{Interpolate}(\boldsymbol{\theta}_{\text{bwd}},\ y)$
    $\theta_{\text{cmd}} \leftarrow \theta_{\text{FF}} - u_{\text{PID}}$
    Apply θ_{cmd} to motor
    Log $\{t,\ F,\ y,\ x,\ \psi,\ v\}$
  end while

  Stop robot; reset PID integral term
end for